Counting Sequences

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to count the number of sequences of length n where each element is an integer between $1 \ldots k$ and each integer between $1 \ldots k$ appears at least once in the sequence.

For example, when $n = 6$ and $k = 4$, some valid sequences are $[1, 3, 1, 4, 3, 2]$ and $[2, 2, 1, 3, 4, 2]$.

Input

The only input line has two integers n and k.

Output

Print one integer: the number of sequences modulo $10^9 + 7$.

Constraints

- $1 \le k \le n \le 10^6$

Example

Input	Output
`6 4`	`1560`